

EXPECTED VALUE, VARIANCE, COVARIANCE, AND CORRELATION

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I NEED A SUMMARY!

In Article No.11, we introduced random variables and probability distributions as mathematical tools for modelling uncertainty. While a distribution describes all possible outcomes of a random variable and their associated probabilities, it is often useful to summarise this information using a small number of key statistics, as if you were typing "summary()" in R. In particular, we may wish to know the average outcome we expect to observe, the degree of uncertainty surrounding that outcome, or how two random variables move relative to one another.

Three of the most important measures used to answer these questions are the expected value, variance, and covariance. The expected value provides a measure of the centre of a distribution, variance quantifies its spread, and covariance measures the relationship between two random variables. Together, these quantities form the foundation of statistical analysis, where they are indispensable in estimating returns and risks of any investments.

EXPECTED VALUE, VARIANCE, COVARIANCE, AND CORRELATION

In Article No.11, we defined the mathematical definitions of expectation and variance. Let us now develop an intuitive understanding of these concepts. The **expected value**, or expectation, of a random variable provides a measure of its long-run average. Intuitively, it represents the value we would expect to observe on average if an experiment were repeated infinitely many times under identical conditions. While the expected value describes the centre of a distribution, it tells us nothing about the degree of uncertainty surrounding that value. Two random variables may have the same expected value but exhibit very different levels of risk. To measure this dispersion, we introduce the concept of variance. **Variance** measures the average squared distance of a random variable from its expected value. A larger variance indicates that outcomes are more widely spread around the mean.

Covariance measures the extent to which two random variables move together and is defined as $Cov(X, Y) = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y]$. Closely related is the **Correlation** coefficient, $\rho(X, Y) = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$. If X and Y are independent, $Cov(X, Y) = 0$ and $\rho(X, Y) = 0$. Do note that the reverse of the correlation relationship need not be true, if $\rho(X, Y) = 0$, then X and Y are said to be uncorrelated, but they may still be dependent.

If $\mathbb{E}[X_i]$ is finite $\forall i = 1, 2, \dots, n$, then $\mathbb{E}[X_1 + \dots + X_n] = \mathbb{E}[X_1] + \dots + \mathbb{E}[X_n]$. The relationship holds whether the X_i 's are independent of each other or not. If X and Y are independent, then $\mathbb{E}[g(X)h(Y)] = \mathbb{E}[g(X)]\mathbb{E}[h(Y)]$. There are also a few rules to calculate variance and covariance. You may have seen the identity $Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$, which more generally, can be rewritten as $Var\left(\sum_{i=1}^n X_i\right) = \sum_{i=1}^n Var(X_i) +$

$2 \sum_{1 \leq i < j \leq n} Cov(X_i, X_j)$. For covariance, we have $Cov\left(\sum_{i=1}^n a_i X_i, \sum_{j=1}^m b_j Y_j\right) = \sum_{i=1}^n \sum_{j=1}^m a_i b_j Cov(X_i, Y_j)$.

When conditioning on new information, it is often useful to summarise the resulting distribution through its conditional expectation and variance. For discrete distributions, we have that $\mathbb{E}[g(X) | Y = y] = \sum g(x)p_{X|Y}(x | y) = \sum_x g(x)p(X = x | Y = y)$. For continuous distributions, we have that $\mathbb{E}[g(X) | Y = y] = \int_{-\infty}^{\infty} g(x)f_{X|Y}(x | y)dx$.

Conditional expectation leads to one of the most useful results in probability theory, the **Law of Total Expectation**, sometimes referred to as the tower property. Thus, for discrete distributions, we have that $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]] = \sum_y \mathbb{E}[X | Y = y]p(Y = y)$. And for continuous distributions, we have $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]] = \int_{-\infty}^{\infty} \mathbb{E}[X | Y = y]f_Y(y)dy$.

AN EXAMPLE OR TWO

Consider two stocks, A and B , whose returns over the next year depend on the overall state of the market. Suppose that there are only two possible market states: a bull market, which occurs with probability 0.6, and a bear market, which occurs with probability 0.4. In a bull market, stock A is expected to return 20% and stock B 10%. In a bear market, stock A is expected to return -10% and stock B -5%. What is the expected return, variance, covariance, and correlation of the two stocks?

Solution: The calculations are straightforward. For stock A , the expected return is $\mathbb{E}[A] = (0.6)(0.2) + (0.4)(-0.1) = 0.08$, while for stock B , we obtain $\mathbb{E}[B] = (0.6)(0.1) + (0.4)(-0.05) = 0.04$. The variances are given by $\text{Var}(A) = (0.6)(0.2 - 0.08)^2 + (0.4)(-0.1 - 0.08)^2 = 0.0216$ and $\text{Var}(B) = (0.6)(0.1 - 0.04)^2 + (0.4)(-0.05 - 0.04)^2 = 0.0054$. We can now calculate the covariance as $\text{Cov}(A, B) = (0.6)(0.2 - 0.08)(0.1 - 0.04) + (0.4)(-0.1 - 0.08)(-0.05 - 0.04) = 0.0108$. This positive covariance indicates that the two stocks tend to move in the same direction across market states. Finally, the correlation coefficient is $\rho(A, B) = \frac{0.0108}{\sqrt{0.0216 \times 0.0054}} = 1$, meaning that the two stocks are perfectly positively correlated. Although stock A exhibits a higher expected return, it also carries a higher level of risk, as reflected by its larger variance.

Suppose that you roll a dice. For each roll, you are paid the face value. If a roll gives 4, 5 or 6, you can roll the dice again. Once you get 1, 2 or 3, the game stops. What is the expected payoff of this game?

Solution: Here we can utilise the law of total expectation. The payoff will be different depending upon the outcome of the first roll. Let $\mathbb{E}[X]$ be the expected payoff and Y be the outcome of the first throw. There is a probability of $\frac{1}{2}$ to get $Y \in \{1, 2, 3\}$, in which case the expected value is the expected face value of 2. That is, $\mathbb{E}[X | Y \in \{1, 2, 3\}] = 2$. There is also a probability of $\frac{1}{2}$ to get $Y \in \{4, 5, 6\}$, in which case it is the expected face value of 5 and any extra throw(s). The extra throw(s) mean that you must restart the game again and have an extra expected value of $\mathbb{E}[X]$. Thus, $\mathbb{E}[X | Y \in \{4, 5, 6\}] = 5 + \mathbb{E}[X]$. Therefore, $\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X | Y]] = \frac{1}{2} \times 2 + \frac{1}{2} \times (5 + \mathbb{E}[X]) \Rightarrow \mathbb{E}[X] = 7$.

CONCLUSION

Expected value, variance, covariance, and correlation are among the most important concepts in probability and statistics, providing concise ways to describe the behaviour of random variables. While the expected value measures the average outcome of a distribution, variance quantifies the uncertainty surrounding that outcome, and covariance and correlation capture the relationships between multiple random variables. These concepts form the mathematical foundation of risk analysis, portfolio construction, and financial modelling. By understanding how to measure both uncertainty and dependence, we can be better equipped to analyse complex problems and make informed decisions in the presence of risk.

ALEXANDER FLANN